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Comparison of ambulatory and polysomnographic recording of jaw muscle activity during sleep in normal subjects

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SUMMARY Clinicians and investigators need a simple and reliable recording device to diagnose or monitor sleep bruxism (SB). The aim of this study was to compare recordings made with an ambulatory electromyographic telemetry recorder (TEL-EMG) with those made with standard sleep laboratory polysomnography with synchronised audio-visual recording (PSG-AV). Eight volunteer subjects without current history of tooth grinding spent one night in a sleep laboratory. Simultaneous bilateral masseter EMG recordings were made with a TEL-EMG and standard PSG. All types of oromotor activity and rhythmic masseter muscle activity (RMMA), typical of SB, were independently scored by two individuals. Correlation and intra-class coefficient (ICC) were estimated for scores on each system. The TEL-EMG was highly sensitive to detect RMMA (0·988), but with low positive predictive value (0·231) because of a high rate of oromotor activity detection (e.g. swallowing and scratching). Almost 72% of false-positive oromotor activity scored with the TEL-EMG occurred during the transient wake period of sleep. A non-significant correlation between recording systems was found (r = 0·49). Because of the high frequency of wake periods during sleep, ICC was low (0·47), and the removal of the influence of wake periods improved the detection reliability of the TEL-EMG (ICC = 0·88). The TEL-EMG is sensitive to detect RMMA in normal subjects. However, it obtained a high rate of false-positive detections because of the presence of frequent oromotor activities and transient wake periods of sleep. New algorithms are needed to improve the validity of TEL-EMG recordings.

KEYWORDS: rhythmic masticatory muscle activity, sleep bruxism, oromotor activity, ambulatory recording, telemetry, electromyogram, polysomnography

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Introduction

Sleep bruxism (SB) is characterised by repetitive jaw muscle activity with grinding or clenching of the teeth during sleep. It is classified as a 'sleep-related movement disorder' in the International Classification of Sleep Disorders (1). For years, SB diagnosis has been based on the patient's awareness of self-generated grinding sounds, the presence of tooth wear and pain or complaints of morning jaw muscle pain or fatigue. However, reliable methods are needed to assess SB and its reduction through various management strategies (2, 3).

Traditionally, SB has been clinically assessed based on reports of tooth-grinding noise and the presence of tooth wear. However, clinicians are recommended to exercise caution when diagnosing SB based on these outcomes. Reports of tooth-grinding noise may not be accurate, as they rely mainly on reports by a sleep partner, although many individuals sleep alone or with

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a sleep partner who remains undisturbed by such sounds. The presence of tooth wear may help distinguish SB patients from normal patients. Again, caution is advised, as the apparent damage may have occurred months or years before clinical observation. Moreover, SB frequency in a given night may be low, moderate or high across patients, and the severity cannot be determined by wear extent in young adults (4, 5). Extensive tooth loss may be due to factors such as the cumulative effects of time, low enamel density or low saliva content, which may contribute to accelerate tooth substance loss (6–8).

Classically, SB is quantified in a sleep laboratory by scoring oromotor activity frequency using polysomnography (PSG) recordings of brain, breathing, muscle and heart activity. More specifically, masseter and/or temporal jaw muscle activity is recorded simultaneously with audio-visual signals. This helps identify transient periods of arousal or awakening during sleep and exclude co-morbidities such as breathing and/or periodic limb movement disorders (2, 9). SB is detected by the presence of rhythmic masticatory muscle activity (RMMA) and is distinguished from other types of oromotor activities such as swallowing and scratching, which are difficult to exclude without audio-visual recordings (5, 9–11). Up to 30% of all jaw muscle activity recorded in SB patients is unrelated to SB. In controls without history of tooth grinding, 85% of jaw muscle activity is unrelated to RMMA (11). Nevertheless, sleep laboratory PSG has several drawbacks. It is performed in an unnatural setting, and scoring is time-consuming and expensive. It is therefore less suitable for large sample studies. Moreover, it is difficult for patients living in remote or rural areas to access PSG (12, 13). Clinicians need a simple, reliable method to assist in SB diagnosis and to assess the efficiency and safety of various SB management approaches.

Attempts have been made to monitor SB-related EMG activity in natural home settings using ambulatory recorders (14–21). Despite the obvious benefits of such devices (lower cost, home monitoring), the specificity of SB motor activity assessment remains a limitation (22). As explained previously, in the absence of simultaneous audio-visual recordings, it is difficult to exclude the presence of non-SB-specific oro-facial movements during sleep (11). Recent commercial multichannel ambulatory PSG systems allow home recordings with audio-visual recording, but their validity remains to be assessed. So far, sleep laboratory PSG with simultaneous audio-visual recording (PSG-AV) is the gold standard for SB quantification in the presence of complex medical conditions (1, 22) or for research purposes (2, 4, 10, 23–27).

To monitor sleep masseter EMG activity with less burden on patients, an ultraminiature cordless masseter EMG measurement system was developed (28, 29). The objective of the present study was to assess the sensitivity, positive predictive value and reliability of RMMA scoring using a single-channel telemetry electromyographic (TEL-EMG) system for recording masseter muscle activity compared with a standard PSG-AV system.

Methods

Subjects

Eight volunteer subjects without apparent current history of tooth grinding (four women and four men; mean age ± s.d. = 26·9 ± 10·9 years; range = 17–49 years) participated in this study. All participants provided their informed consent in accordance with the institution's ethical rules (Hôpital du Sacré-Coeur de Montréal).

Telemetry EMG recording device (TEL-EMG)

A BMS TEL-EMG recorder* was used (Fig. 1). This telemetry device consists of an ultraminiature transmitter unit and a receiver unit (28, 29). The transmitter unit contains 12-mm-diameter bipolar electrodes, a reference electrode, a 500 gain amplifier, 10–500 Hz bandpass filters, a 1-kHz analog-to-digital (A/D) converter at a 1-kHz sampling frequency, an 8-bit central processing unit (CPU: HD64F38076R;†), a battery and a transmitter. The transmitter unit is 42·5 × 37·0 × 6·5 mm in size and weighs 12·1 g with the battery. Amplified EMG digital data are transmitted to the receiver unit. The obtained EMG data are stored in a compact flash memory card (CF card), and the card is then transferred to a personal computer for analysis. The transmitter unit was attached over the masseter muscle by three adhesive gel pads on electrodes to conduct electrical signals (one side, randomised among subjects). The receiver antenna was placed beside a pillow.

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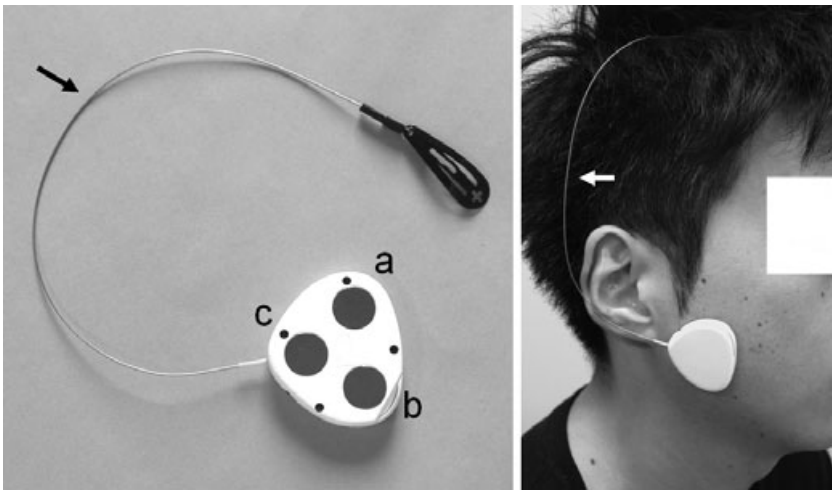

Fig. 1. Photographs of the transmitter unit of telemetry electromyographic. The arrow indicates the fine and flexible antenna of the transmitter unit. Bipolar electrodes (a, b) for recording and a reference electrode (c) are positioned in triangle.

PSG audio-visual recordings (PSG-AV)

The polygraphic recordings, using surface electrodes, included electroencephalograms (EEG; C3A2; O2A1 positions), bilateral electrooculograms (EOG), electrocardiograms (ECG) and electromyograms (EMG) from the chin, masseter (one side, opposite of the TEL-EMG site, randomised among subjects), bilateral temporalis and tibialis muscles. All signals were amplified and recorded at a 256-Hz sampling rate and stored for offline analysis using Harmonie Software[‡]. Audio-visual recordings were performed simultaneously.

Masseteric EMG and sleep scoring

Subjects spent one night in a sleep laboratory. Simultaneous EMG recordings were made with a TEL-EMG attached to one masseter muscle and a PSG electrode attached to the opposite masseter. TEL-EMG and PSG-AV masseter recording side was randomised among subjects. TEL-EMG was on the left side for six subjects and on the right side for the other two subjects.

Five tapping movements before sleep and after getting up in the morning were performed, and the first burst of the tapping movements was used for synchronisation between the PSG recording system and the TEL-EMG device.

Sleep stage was scored according to standard criteria (30, 31). Masseter and temporal EMG bursts with duration exceeding 0·25 s were selected for oromotor activity scoring, according to published criteria (11, 32). Cut-off value of amplitude was twice the baseline activity, and the bursts with greater amplitude than the value were selected. Thus, oromotor episodes (OME) separated by 3-s intervals were recognised as RMMA if they corresponded to one of the three following patterns: phasic (three or more masseter-temporalis EMG bursts, each lasting 0·25–2·0 s), tonic (one masseter-temporalis EMG burst >2·0 s) or mixed (both masseter-temporalis burst types) episodes (11, 32). Both scoring on the TEL-EMG recording and the PSG were manually done. EMG waves of TEL-EMG and PSG were displayed on computer monitors. EMG bursts with amplitude greater than twice the baseline activity were detected visually on the monitor. When the wave size was around twice the baseline activity and visual discrimination is difficult to decide, the value of amplitude of the burst was checked to confirm whether the burst is greater than twice the baseline activity or not.

Rhythmic masseter muscle activity were also scored using PSG and audio-visual recordings (PSG-AV). Other types of oro-facial activity were also identified in PSG-AV signals. Swallowing, coughing and myoclonus were identified as well as other movements such as face and head scratching (12, 33, 34). Oromotor activity during wakefulness was excluded from PSG-AV scoring.

Rhythmic masseter muscle activity-related oromotor episodes detected by PSG-AV were used as the reference to validate those observed in TEL-EMG recordings. Episodes scored on the TEL-EMG were considered a true oromotor episode (T-OME) when RMMA was observed with both recording systems during sleep. Many non-RMMA-specific OME were observed and considered other oromotor episodes (O-OME). O-OME can occur during wakeful periods of sleep (O-OME/

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wake) or during sleep (O-OME/sleep), as confirmed by PSG. Note that because no subjects in this study were sleep bruxers (tooth grinders), the data must be extrapolated with caution.

Only one author (TY) scored the TEL-EMG signals, and only one masseter EMG channel was scored, with no knowledge of the PSG-AV scoring data. An experienced sleep technician scored the PSG-AV signals blind to the scoring with TEL-EMG. Afterwards, the scoring data were matched between TEL-EMG and PSG-AV.

Data analysis and statistics

Telemetry-EMG OME and T-OME were compared and grouped into true-positive, false-positive and false-negative episodes. The sensitivity and positive predictive value of TEL-EMG detection were calculated (35). Bland–Altman plots were used to graphically display the difference between the number of oromotor episodes detected by the TEL-EMG and by PSG-AV (36). Correlation between TEL-EMG OME and T-OME was estimated using the Spearman's rank correlation coefficient. The intra-class correlation coefficient (ICC) was also calculated to estimate reliability (37).

Results

Figure 2 illustrates an oromotor episode, classified as T-OME, using the TEL-EMG. This was detected during sleep by both the TEL-EMG and PSG, showing matching

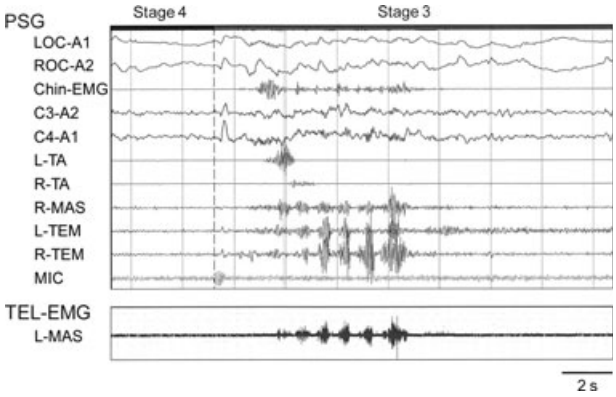

Fig. 2. Sleep recordings by polysomnography-AV (upper panel) and telemetry electromyographic (lower panel). LOC-A1 and ROC-A2, left and right electrooculogram; C3-A2 and C4-A1, electroencephalograms; L-TA and R-TA, left and right tibialis muscles; L-MAS and R-MAS, left and right masseter muscles; L-TEM and R-TEM, left and right temporalis muscles; MIC, microphone.

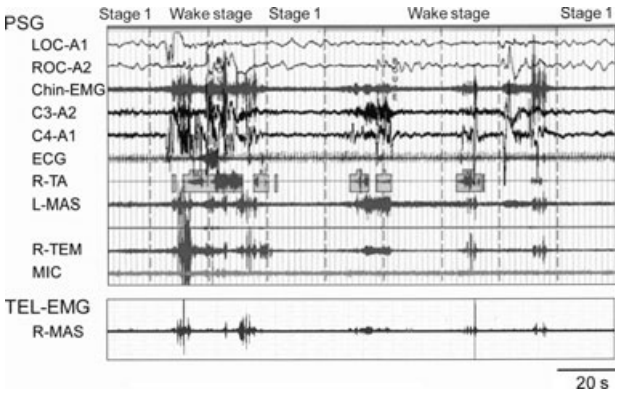

Fig. 3. Wake period during sleep with rise in muscle activity non-specific to rhythmic masseter muscle activity oromotor episode called O-OME/wake. Upper panel, polysomnography-AV; lower panel, telemetry electromyographic. LOC-A1 and ROC-A2, left and right electrooculogram; C3-A2 and C4-A1, electroencephalograms; ECG, electrocardiograms; R-TA, right tibialis muscles; L-MAS and R-MAS, left and right masseter muscles; R-TEM, right temporalis muscles; MIC, microphone.

EMG patterns at a given time using both recording methods. An oromotor episode detected by the TEL-EMG and occurring during wakefulness, classified as O-OME/wake using the TEL-EMG, is shown in Fig. 3.

During wake and sleep periods, a mean of 44·9 OME per subject was detected using the TEL-EMG compared with 10·5 RMMA episodes per subject using PSG-AV (Table 1). TEL-EMG sensitivity during wake and sleep periods was 0·988, but with a positive predictive value of only 0·231. Of the total TEL-EMG OME, the false-positive rate was 76·9% (Fig. 4). A large percentage (73·0%) of TEL-EMG OME was classified as O-OME compared with PSG-AV, whereas only a small percentage (3·9%) of false-positive TEL-EMG OME was undetected using PSG-AV. O-OME identified using the TEL-EMG during sleep (O-OME/sleep) were for usual sleep-related oromotor activity such as swallowing, face scratching and other (Table 2). In addition, 71·7% of false-positive TEL-EMG OME occurred during wake periods (O-OME/wake). Therefore, when RMMA matching was limited to episodes during sleep, the positive predictive value of sleep-period TEL-EMG OME increased (0·516, Fig. 4 and Table 1).

Bland–Altman plots showed a large discrepancy between the number of oromotor episodes detected using the TEL-EMG and using PSG-AV during wake periods of sleep and during uninterrupted sleep. The plots converged to a relatively narrower area, i.e. this difference decreased, when the comparison was limited

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Table 1. Scoring data of oromotor episodes

		Wake and sleep periods	Sleep period	Wake period
T-OME	Mean	10·5	10·5	–
	s.d.	20·7	20·7	–
T-OME/h	Mean	1·6	1·6	–
	s.d.	3·2	3·2	–
BMS-OME	Mean	44·9	20·1	24·8
	s.d.	32·2	24·4	11·2
True positive	Mean	10·4	10·4	–
	s.d.	20·7	20·7	–
False positive	Mean	34·5	9·8	24·8
	s.d.	15·1	6·3	11·2
False negative	Mean	0·1	–	–
	s.d.	0·4	–	–
Sensitivity		0·988	0·988	–
Positive predictive value		0·231	0·516	–

Mean of oromotor episodes was calculated per subject.
Sensitivity and positive predictive value of BMS were calculated using T-OME as the standard.
Sensitivity = true-positive episodes/(true-positive episodes + false-negative episodes).
Positive predictive value = true-positive episodes/(true-positive episodes + false-positive episodes).
T-OME, true oromotor episode detected with polysomnography (PSG) audio–video recording; T-OME/h, T-OME per hour of sleep; BMS-OME, oromotor episode detected with BMS recording; s.d., standard deviation.

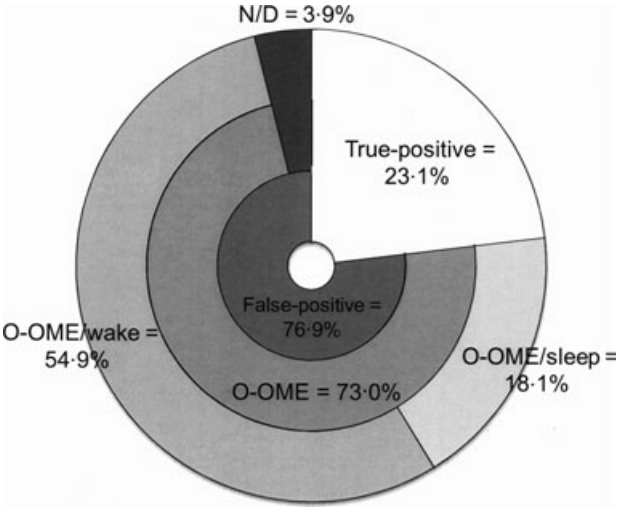

Fig. 4. Distribution of oromotor episodes recorded with the telemetry electromyographic (TEL-EMG). Total number of 359 oromotor episodes with the TEL-EMG from the eight subjects is divided into categories by comparison with polysomnography (PSG) scoring. O-OME, other oromotor episodes; O-OME/wake, other oromotor episodes during wake period; O-OME/sleep, other oromotor episodes during sleep; N/D, undetected with PSG-AV recording.

Table 2. Number of other oromotor episodes during sleep periods and detailed contents

	Mean number per subject	s.d.	
OME/sleep	8·1	4·2	100·0%
Contents			
Lip sucking	0·1	0·4	1·5%
Chewing-like	1·6	2·2	20·0%
Swallowing	1·3	1·8	15·8%
Coughing	0·1	0·4	1·5%
Talking	0·3	0·7	3·1%
Other mouth movement	1·4	2·1	16·9%
Eye opening/blinking	0·4	0·7	4·6%
Face scratching	4·1	3·6	50·8%
Head movement	4·6	3·8	56·9%
Other body movement	3·5	2·1	43·1%

Total of each percentage is larger than 100% due to overlap.
O-OME/sleep, other oromotor episode during sleep periods; s.d., standard deviation.

to episodes during sleep (Fig. 5). The Spearman's rank correlation coefficient revealed a non-significant correlation between TEL-EMG OME and T-OME (ρ = 0·488, P = 0·197) during wake periods of sleep and during sleep. During uninterrupted sleep, the correlation was also non-significant (ρ = 0·605, P = 0·109, Fig. 6). The ICC was low (0·47) when wake episodes during sleep were considered. However, after removing the influence of wake episodes during sleep, the ICC was higher during uninterrupted sleep (0·88), indicating good reliability.

Discussion

To our knowledge, this is the first study to assess the sensitivity, positive predictive value and reliability of an ambulatory device for masseter muscle recording compared to PSG with simultaneous audio-visual recording as the reference. The results demonstrated that the TEL-EMG device was highly sensitive to detect masseter muscle activity, but showed low positive predictive value for RMMA episode detection, typical of SB activity. This was because most OME identified by the TEL-EMG occurred in wake periods (O-OME/wake) or were not specific to RMMA (e.g. swallowing and scratching, O-OME/sleep). These findings suggest that the TEL-EMG device is reliable in detecting masseter muscle activity, although corrections must be made before interpreting data collected in the absence of a method to assess wake episodes during sleep.

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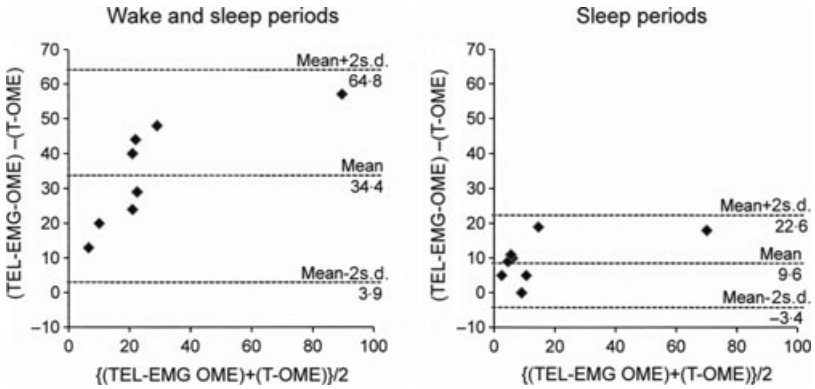

Fig. 5. Bland–Altman plots of the differences between oromotor episodes detected with the telemetry electromyographic (TEL-EMG OME) and true oromotor episodes (T-OME) detected with polysomnography-AV. Mean and ± 2 standard deviations (s.d.) are shown.

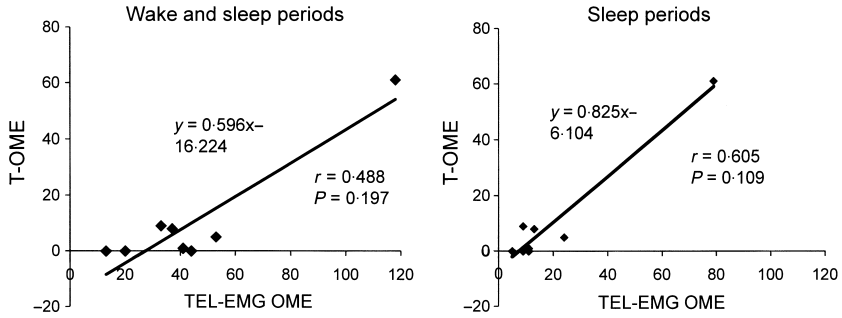

Fig. 6. Correlation between oromotor episodes detected with the telemetry electromyographic (TEL-EMG OME) and true oromotor episodes (T-OME) detected with polysomnography-AV.

Based on a previous work, it appears that sleep recordings are more prone to false-positive detections than wake recordings. In a recent publication, our laboratory showed that in control subjects, 85% of OME scored using PSG-AV are not specific to the RMMA that usually characterise SB (11). In a previous study, one of the authors also found that daytime validity of the TEL-EMG, based on distinct EMG burst characteristics (e.g. tooth tapping, maximum tooth clenching and gum-chewing), was comparable to conventional polygraphic day wake time recordings (28). In addition, motion artefacts of signals during day wake time TEL-EMG measurements were equivalent to or less than those during daytime polygraphic measurements (29). The fairly good sensitivity obtained in the present nocturnal study may be attributed to the accurate EMG recording properties of the TEL-EMG. A better positive predictive value was found when wake periods were excluded from the calculation. This suggests that methods to improve the detection of wake periods during sleep and/or non-RMMA OME may improve the validity of limited-channel recording systems for RMMA-SB studies, as discussed later in the text.

Oromotor activities are usually scored only during sleep in PSG-AV studies. However, subjects between 20 and 40 years of age are awake after sleep onset for an average of 20 min (10, 38, 39). Characterising these oromotor activities in the wake periods that follow sleep onset could be made to better understand SB and validate ambulatory recordings.

Other ambulatory recorders have been developed (14–21). However, very few studies have compared ambulatory recording to the gold standard, PSG-AV, in assessing the specificity of masseter muscle activity in relation to SB-related RMMA. In a previous study (17), bioelectric signals from a male bruxer were recorded for one night simultaneously with both an ambulatory recorder and polysomnography. Results showed good agreement (sensitivity) with the detection of bruxism episodes by the masseter electromyogram, as well as with the results of the present study. However, in that benchmark study, the number of false-positive episodes was not mentioned, and positive predictive value and specificity were not estimated. Previous sleep laboratory PSG studies have shown that other oro-facial activities during sleep, such as swallowing, coughing and myoclonus, are not specific to RMMA-SB and should be identified as other oro-facial activities (11, 33). These oro-facial motor activities during sleep, scored based on simultaneous and synchronised audio-visual recordings, included swallowing, coughing, face rubbing, body movements, lip sucking, head movements, chew-

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ing-like movements, head rubbing, face scratching, eye opening and blinking. From these findings, it was predicted that SB assessment using ambulatory EMG recordings only would potentially include a considerable number of oromotor sleep activities, which are false positives. The present study demonstrated that, notwithstanding its sensitivity, a single-channel EMG device detected too many non-specific oromotor activities that were identified as false-positive episodes (O-OME in the present study). A specificity analysis requires accurate data on the number of true-negative episodes. As it was difficult to define a true-negative episode under the present research design, a valid calculation of TEL-EMG specificity was not possible. However, the large number of false-positive episodes indicates low specificity for SB only using the ambulatory EMG device.

In an SB clinical study or research trial, researchers may observe concomitant sleep disorders such as PLMS (identified with EMG of the anterior tibialis muscle), snoring and sleep apnoea/hypopnoea syndrome (SAHS) and, more rarely, epileptic sleep activity (RBD) (22, 27, 40). It should be noted that single-channel recording does not allow recognition of concomitant sleep disorders. PSG-AV is therefore mandatory in the presence of complex medical or sleep disorders.

To counter the above-mentioned disadvantage of TEL-EMG recording, simultaneous use of another channel (heart rate or EEG) can be used to help detect sleep disorders. Recently, ambulatory sleep recognition devices (41, 42) have been developed. In the present study, when wake period episodes were eliminated using PSG sleep assessment, the ICC was 0·884, indicating good reliability. Simultaneous use of an adequate ambulatory sleep recognition device and a TEL-EMG device plus sleep and sleepiness questionnaires (Epworth, sleep inventory) may be an alternative method to reduce false-negative episodes. Furthermore, an algorithm could be developed for TEL-EMG scoring to allow discriminating between T-OME and O-OME. For example, the present TEL-EMG scoring used a data threshold that was more than twice the amplitude of the baseline activity. At a smaller threshold (e.g. the 10–20% used for the PSG RMMA scoring), small O-OME bursts could be included and the rate of false-negative episodes could be higher. A higher threshold for burst amplitude over burst duration or other criteria would probably be needed for single-channel EMG recording of masseter activity in the absence of audio-visual recording. A recent innovation includes an algorithm for a time–frequency analysis with discriminative criteria (43). Furthermore, an actigraphic movement sensor could be used concomitantly (44). Such device might improve wake detection if the time window is sufficiently large to allow discriminating between active and quiet periods during the total time the patient is in bed sleeping. The above-mentioned countermeasures should effectively improve the validity of the TEL-EMG's quantitative estimation of oromotor activities during sleep.

While the present study provided new important information, several limitations should be noted. First, the sample size of this study was small and the subjects were not sleep bruxers. Normal subjects have a low incidence of RMMA episodes per night in addition to all the other physiological oro-facial activities that occur during sleep. The lower incidence of RMMA in normal subjects leads to a speculation that in sleep bruxers, the ratio of O-OME to T-OME will be smaller than in normal subjects, and positive predictive value will be larger than the value obtained in the present study. We should take the possible difference between sleep bruxers and normal subjects into consideration when we interpret the result of the present study. Second, we did not have independent scoring of the control masseteric and temporal EMG trace of PSG blind to the other traces of PSG such as EEG and AV data. If we had the data, further insights into the intrinsic quality of the new portable recorder could be given. We think that such analysis is further assignment in future study. Third, the cut-off value of burst amplitude used in this study was only one (i.e. twice the baseline activity). Use of several levels of cut-off value concerning burst amplitude, burst duration and other item of burst characteristics could give further information to develop new effective algorithms for eliminating non-specific oromotor activities. To improve the validity of ambulatory EMG recordings, simultaneous use of an adequate ambulatory sleep recognition device and/or an adequate movement sensor with well-designed burst recognition algorithms is needed. In addition, further comparative studies with larger sample of sleep bruxers are needed between ambulatory EMG and PSG-AV.

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